

Anomaly Detection using Minimum Covariant Determinant as Feature in Multivariate Data

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ABSTRACT

Ensuring reliability in the data generated by various applications enables data analyst to take accurate decisions. But the presence of anomalous or faulty readings may affect the accuracy of decision making. Many statistical and machine learning techniques have been developed by research community and widely used for anomaly detection. To this end, this paper presents a method that leverages minimum covariance determinant based mahalanobis distance to improve the feature set and then refined feature set is supplied to various machine learning algorithm. The performance of the proposed approach has been evaluated on various machine learning algorithms including probabilistic model, statistics based binary classifier, Isolation based model and neural network-based model. It is observed from the results that the proposed approach reasonably improves the machine learning methods performance metrics such as detection accuracy, precision, recall and F1 score of the machine learning models.

CCS CONCEPTS

• **Computing methodologies**; • **Machine learning**; • **Machine learning approaches**; • **Classification and regression trees**; **Neural networks**; • **Security and privacy**; • **Intrusion/anomaly detection and malware mitigation**;

KEYWORDS

Mahalanobis Distance, Minimum Covariant Determinant

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1 INTRODUCTION

“Anomaly is defined as an observation (or subset of observations) which appears to be inconsistent with the remaining data” [1]. The detection of anomalies has significant importance as it has a variety of applications ranging from network intrusion detection, medical and public health, credit card fraud detection, forest fire detection,

industrial applications, image processing, sensor networks to many others [2]. Anomaly Detection is also referred as outlier detection. Several researchers have attempted to solve anomaly detection problem using machine learning (ML) techniques. ML offers many advantages such as fast data analysis, self-learning, fault diagnosis and prediction along with identification of trends.

The machine learning algorithms are trained to accept input and predict its corresponding output. If the machine learning algorithms are not provided with sufficient input information and proper hyper parameters these algorithms often under-perform for anomaly detection [3]. On the other hand, it has been demonstrated by various researches that the distance of a point is an effective means for detecting the anomalies [4]. Outliers are determined based on the ‘closeness’ of the data points using a suitable distance measure [5]. Points which lie away from the other data points are more susceptible to be outliers. We can use the distance information to improve the input data provided to the ML model to detect anomalies more accurately.

Mahalanobis distance is one of the widely used multivariate distance measure for anomaly detection. It considers the shape and size of multivariate data by means of covariance matrix. In mahalanobis distance the distance between two points or a point and remaining sample distribution is normalized with the help of the covariance between the points or point and sample. Mahalanobis distance has been effectively used by various researchers for anomaly detection problem [6, 7] but mahalanobis distance suffers from a drawback. The mean of the sample (μ) and covariance (Σ) between point and sample can drift away arbitrarily far even by the presence of single bad outlier. This is known as masking effect.

To avoid masking problem, Rousseeuw et al., proposed Minimum Covariant Determinant (MCD) [7]. MCD based mahalanobis distance is a robust variant of mahalanobis distance which avoids the effect of outliers [8]. MCD based Mahalanobis distance is shown to be more effective for detection of outliers [8]. Although many researchers have utilized the concept of MCD for anomaly detection and application of MCD for improving accuracy of detecting anomalies is widely practiced by researchers in various fields. MCD has been used widely for simplifying problems or estimating cut off to separate outliers [9] or as a distance measure in larger models [10]. However, the availability of algorithms which utilizes the concept of MCD based mahalanobis distance for improving feature set is limited. This has drawn our attention to attempt such research problem. This paper presents a unique approach for anomaly detection by improving the original features of the data by means of Minimum Covariant Determinant based mahalanobis distance. Our conclusion is that the proposed approach performs consistently better on different types of machine learning techniques (i.e., supervised/ unsupervised) over data sets of varying sizes and type. This shows the potential applicability of our proposed approach.

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In this paper, we have focused on the problem of anomaly detection from both supervised learning and unsupervised learning perspective and studied the performance of method using various performance parameters which are Accuracy, Precision, Recall, F1 Score AUC ROC, and AUC PR by the use of labeled datasets. We have also studied the effect of proposed approach on the training time and testing time. Following different types of models have been considered to evaluate the proposed approach.

- Probabilistic Model (GMM clustering)
- Statistic Based Binary classification model (Logistic Regression)
- Isolation Based (Isolation Forest)
- Neural Network based (Self organizing map/Kohonen Map)

The main contributions of this paper are

- This paper proposes a hybrid approach for detection of anomalies in the data to combine MCD based mahalanobis distance with different types of machine learning models.
- Evaluates the performance of the proposed approach by applying them on different ML models and compares it with the performance of the same model when using regular features. Time required by these models in both the cases has also been compared.

The proposed method introduces a unique framework that incorporates the multivariate location of data points through the utilization of MCD-based Mahalanobis distance, enabling the detection of outliers. By providing additional information in the form of MCD Distance, the method enhances the effectiveness of machine learning algorithms for outlier detection in the given data. The proposed method has demonstrated its compatibility with various types of machine learning algorithms, including probabilistic algorithms, statistics-based binary classifiers, isolation methods, and neural network-based models. It is particularly advantageous for identifying outliers in multivariate data and can effectively handle highly contaminated datasets due to the robustness of MCD as a multivariate location estimator, which mitigates the impact of outliers.

This paper presents a comprehensive study on the impact of the proposed approach by utilizing a range of performance evaluation parameters, and provides an in-depth analysis of the obtained results. Furthermore, the research showcases the potential of employing MCD-based Mahalanobis distance for anomaly detection in conjunction with machine learning algorithms.

2 RELATED WORK

It is worth noticing that distance based methods can be combined with machine learning algorithms for achieving better performance of the machine learning algorithms. An example of this type of fusion is the work carried out by Sarmadi et al. [11]. Anomalies are points whose distance is more than a cutoff value from other points. It has been established in the literature that employing a distances based on robust estimate of the multivariate location of point is advantageous to determine outliers in multivariate data [7]. Several studies have revealed that effectiveness of MCD for outlier detection by various researchers. Some of the significant works are presented in this section.

Hardin et al. [10] have applied MCD based mahalanobis distance for the outlier's detection in multiple cluster setting. The threshold values are computed from distributional quantiles of chi-square test and F distributions of the data. Yu et al. [12] have also attempted outlier detection with minimum covariance determinant for novelty class detection in machine learning-based condition diagnosis. They have used hard voting ensemble using various ML algorithms. Their method utilizes GA and PCA based feature selection technique. Leysa et al. [13] 2017 used MCD based mahalanobis distance for detection of multivariate outlier detection and demonstrated the superiority of MCD over mahalanobis distance. Hussain et al. [14] has also applied MCD with other machine learning models for anomaly detection for crash risk estimation.

To improve the performance of MCD for outlier detection many versions of MCD have been introduced by the research community for the detection of outliers. Rousseeuw et al. [7] introduced FAST-MCD algorithm which optimizes the time-consuming procedure used by MCD to improve the calculation speed of MCD [15]. Although the FAST-MCD algorithm is fastest of all the existing MCD algorithms but the main difficulty is determining its key parameter 'h' (where 'h' is the set of observations for which the determinant of the sample covariance matrix is as small as possible), choice of 'h' plays a vital role in the determination of outlier efficiently. Fast-MCD requires substantial running times for large no of features as the number of candidate solutions grows exponentially with the increasing no of features 'p' of the sample leading to high running time complexity for even smaller problems. Verardi et al. [16] have introduced a relatively fast version of MCD based on 3 step procedures which select only 10 p-subsets with the smallest preliminary determinants instead of creating all they also presented the formula to find minimal number of p-subsets (N) to be considered. Wu et al. [9] Proposed a modified version of MCD i.e. M-MCD which can get the optimal value for parameter h by using the self-adaptive iteration. The performance of M-MCD was demonstrated on chemical process data. In this way MCD has been used vastly in various domains for anomaly detection by the research community, to this end this paper proposes a MCD based feature selection approach for anomaly detection.

3 MATERIALS AND METHODS

3.1 Mahalanobis Distance (MD)

Mahalanobis distance is a multivariate distance measure introduced by Indian statistician P. C. mahalanobis in 1936. Mahalanobis distance gives the idea of point's deviation from the distribution in the setting of Gaussian distributions [17]. MD is used to find the distance between two distributions or the distance between a point and the distribution. "Mahalanobis distance measures the distance of a point relative to the centroid (mean) of a distribution". The squared mahalanobis distance between two populations with means respectively μ_1 and μ_2 and common variance Σ can be defined by following equation [18].

$$MD^2(\mu_1, \mu_2, \Sigma) = (\mu_1 - \mu_2)^T \Sigma^{-1} (\mu_1 - \mu_2) \quad (1)$$

Where, Σ is covariance matrix which is a positive definite matrix by the property of covariance. This implies that MD comes out to be a real number hereby it is free of any unit of measurement. The

shape and size of multivariate data can be figured by the covariance matrix. Mahalanobis distance utilizes the covariance which helps to take the shape and size of multivariate data in account while calculating the outlier-ness [17]. In some cases consideration of shape of distribution is critical for determination of outliers as points which are distant can also be normal points depending on the size of the data distribution. For this reason distance-based methods perform poorly when the data contains clusters of different densities [19] since they only consider the distance of a point from other point but not the position of the point with respect to data distribution. In such cases a distance measure should be preferred which takes the shape of distribution into account such as [13].

3.2 Minimum Covariance Determinant (MCD)

The minimum covariance determinant (MCD) method of Rousseeuw [7] is a highly robust estimator of a location of a point in multivariate space. The classical estimates such as the Mahalanobis distances can be so strongly affected due to masking effect and they can become ineffective to detect the outliers [20]. To avoid the masking effect, robust estimators of multivariate distance are needed to avoid the outlier, if they are present. The results of various research [20, 21] support the idea that minimum covariance determinant (MCD) is such robust method for estimation of points multivariate location and scatter which can resist outliers.

The Concept of MCD is based on the robust distances. While calculating the Mahalanobis distance instead of entire data (which may contain outliers) MCD considers

1. μ_{MCD} , which is the mean of the 'h' observations 'h' is the chosen set of points for which the determinant of the sample covariance matrix is as small as possible;
2. Σ_{MCD} , which is the corresponding covariance matrix.

$$MCD(x) = MD\left(x, \mu_{MCD}, \Sigma_{MCD}\right) \quad (2)$$

MCD is mahalanobis distance using μ_{MCD} and Σ_{MCD} , where μ_{MCD} is the MCD estimate of location and Σ_{MCD} is the MCD covariance estimate. The covariance matrix gives the information about how broad the distribution is i.e., how much the variables "vary" with each other. As anomalies lies away from the rest of the distribution so selecting a tightly distributed subset will possibly exclude the anomalies. The basic idea of MCD is to find h observations out of n samples (where $h < n$) that minimizes the determinant of the covariance matrix is the MCD of those data [7, 10]. It tries identifying the subsample containing half of the observations that is associated with the smallest generalized variance i.e., smallest multivariate spread. To identify such a subsample, the idea is to consider all possible subsets containing 50% of the observations and flag the one with the smallest covariance matrix determinant [22]. Due to its robust practical nature MCD is preferred by researchers in field of analytics to detect outliers. Consistency of the raw MCD estimator of location and scatter at elliptical models has been proved in the literature [21].

3.3 OUTLIER DETECTION USING MACHINE LEARNING

Four different types of machine learning algorithm have been considered in this work. The selected parametric and nonparametric

algorithms for this study come from a variety of approaches including probabilistic algorithms; statistics based binary classifier, isolation methods and neural networks-based model.

3.3.1 Gaussian Mixture Model (GMM). GMM is a probabilistic model which is a distribution-based clustering. GMM is based on the assumption that all the data points are generated from a mixture of a finite number of Gaussian distributions with unknown parameters [23]. The Gaussian models are used for representing normally distributed subpopulation within an overall population. GMM allows the model to learn the subpopulations automatically. GMM requires specifying the number of components as an input.

GMM model can be used to perform either hard clustering or soft clustering. In hard clustering data point is assigned to multivariate normal component that has highest posterior probability of [24]. Soft clustering (or Fuzzy clustering) is a more flexible form of GMM. In which a score is assigned to a data point respective to each cluster based on the association strength of the data point to the cluster. In soft clustering data point is assigned to more than one cluster. GMM clustering is most suitable for the data which has clusters of different sizes and correlation within them.

3.3.2 Self-Organizing Map (SOM). Self-Organizing Maps (SOMs) is a well-known unsupervised neural network approach to clustering. The SOM (Self Organizing Map) Neural Network or commonly known as Kohonen neural network system classifies the units by the similarity of a particular pattern in the same class. An unsupervised Self-Organizing Map can also be implemented for a classification problem. To implement SOM as a classifier the model is applied on a data without labels and later results are confirmed by projecting the labels on the data. In terms of the computational cost of the algorithm the time complexity depends on the number of iterations, the number of features and the number of neurons [25]. The advantage of SOM algorithm lies in its capacity to handle large datasets efficiently. It is also robust even when the data is noisy and linearly non-separable [26].

3.3.3 Isolation Forest. Isolation Forest also known as iForest is a model that works by isolating the anomalies from normal patterns. It constructs an ensemble of iTree(Isolation Tree) for the data based on the values of its features. Anomalous data have very different attribute values from normal instances so iTress with short average path lengths represent anomalies.

Isolation Forest algorithm recursively generate the partition on the data in such a way that by the end of the process each point is isolated. iForest only requires to build partial model since a large part of an iTree that isolates normal points is not needed to be constructed for anomaly detection. Eventually the execution is fast and memory requirement is low. The performance of isolation forest is better than other algorithm in the domain of anomaly detection [27]. Isolation forest-based anomaly detection approach has been widely used for anomaly detection. Li et al. [28] has used a kernel isolation forest for anomaly detection [29]. Safaei et al. [30] has demonstrated that isolation forest works best among other machine learning models for anomaly detection using different types of correlation of data as features.

3.3.4 Logistic Regression. Logistic regression is a supervised machine learning technique which is based on multivariate regression

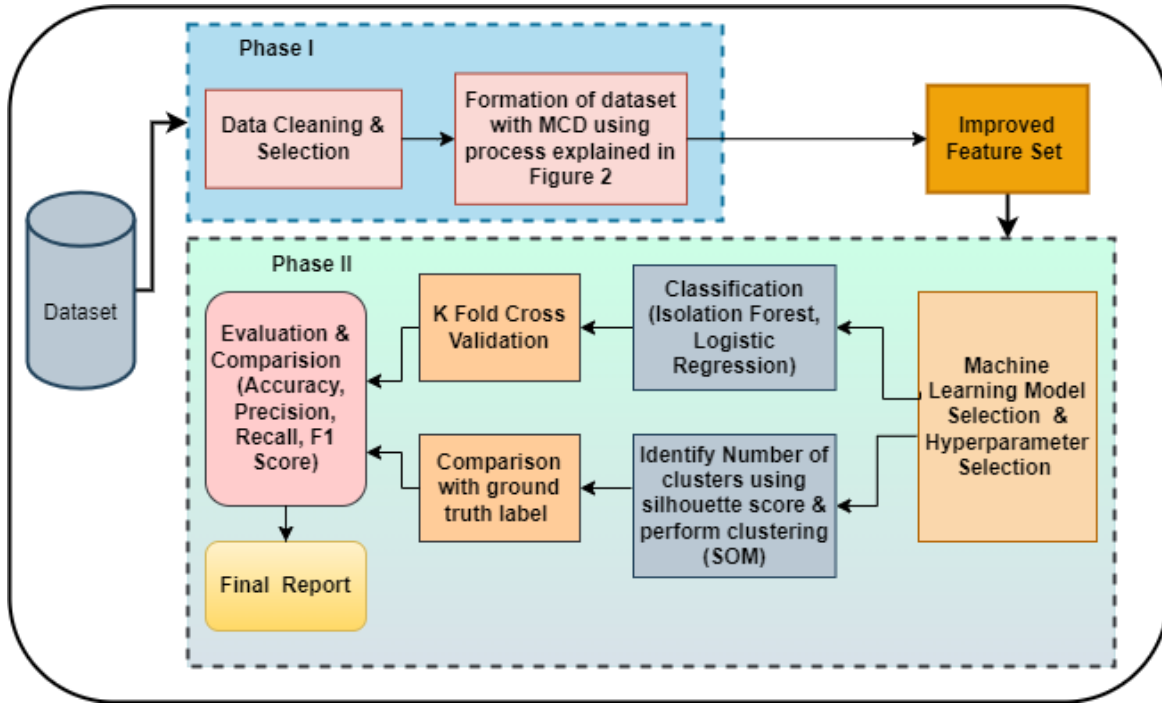


Figure 1: Methodology of Work

analysis to identify the relation between input and output labels. Logistic regression is a statistical methodology which is most suited in cases where predicted outcomes in the data are binary such as positive/negative or anomaly/Normal. Logistic regression is based on the logit transformation of the dependent variable. The logit transformation creates a continuous logarithmic curve from non-continuous data. The model is built on the basis of the outcome probabilities for each dependent variable value [31].

4 METHODOLOGY OF WORK

The proposed anomaly detection methodology is a hybrid approach which utilizes the concept of distance-based anomaly detection to benefit the machine learning techniques. The objective is to find anomalous points from the data. The proposed method works in two phases as shown in Figure 1. Figure 2 represents the phase 1 of the proposed method which forms the new dataset with a new feature based on MCD based mahalanobis distance.

For the formation of modified dataset in phase 1, the data corresponding to dataset is chosen and provided as input for calculation of MCD based mahalanobis distance corresponding to each data point x_i using the following equation

$$\begin{aligned} mcdScore[x_i] &= MCD^2(x_i, \mu_{MCD}, \Sigma_{MCD}) \\ &= (x_i - \mu_{MCD})^T \Sigma_{MCD}^{-1} (x_i - \mu_{MCD}) \end{aligned} \quad (3)$$

Where x_i is a data point, μ is the mean of sample and Σ is the covariance matrix. The calculation of MCD based mahalanobis distance is based on μ_{MCD} and Σ_{MCD} . Where μ_{MCD} is the MCD estimate of Mean of the sample and Σ_{MCD} is the MCD covariance estimate of x_i w.r.t sample. Algorithm 1 presents the proposed approach for

anomaly detection. Table 1 presents the various symbols used and their meaning. The algorithm generates MCD distance of each point from the distribution to create mcdScore. A new dataset is formed by combining mcdScore as a feature value with the other existing features of the data. The basic idea behind this approach is that points which have long distance from other data points are more susceptible to be outlier [32]. By applying machine learning models on newly formed dataset obtained from phase1, the information provided to the machine learning can be improved for the correct classification of the data which leads to better identification of the anomalies.

The new dataset is the combination of all the relevant features from original dataset and the mcdScore respective to each point i.e., MCD distance of each point.

The new dataset is provided along with the other required hyper parameters to various machine learning models. Parameters for machine learning algorithms are tuned manually to select the best parameters. The approach has been tested on both supervised and unsupervised machine learning algorithm. In supervised machine learning algorithms Logistic Regression, Isolation Forest and GMM Classifier is used. Whereas in unsupervised ML algorithm self-organizing map (kohonen Map) clustering has been used. In this work we have run machine learning models with two different input settings.

Setting I- using feature set present in the original dataset referred as “regular features set”.

Setting II- using feature set corresponding to the new dataset resulted from phase1. This feature set is the combination of regular feature set and MCD distance respective to each point. The feature

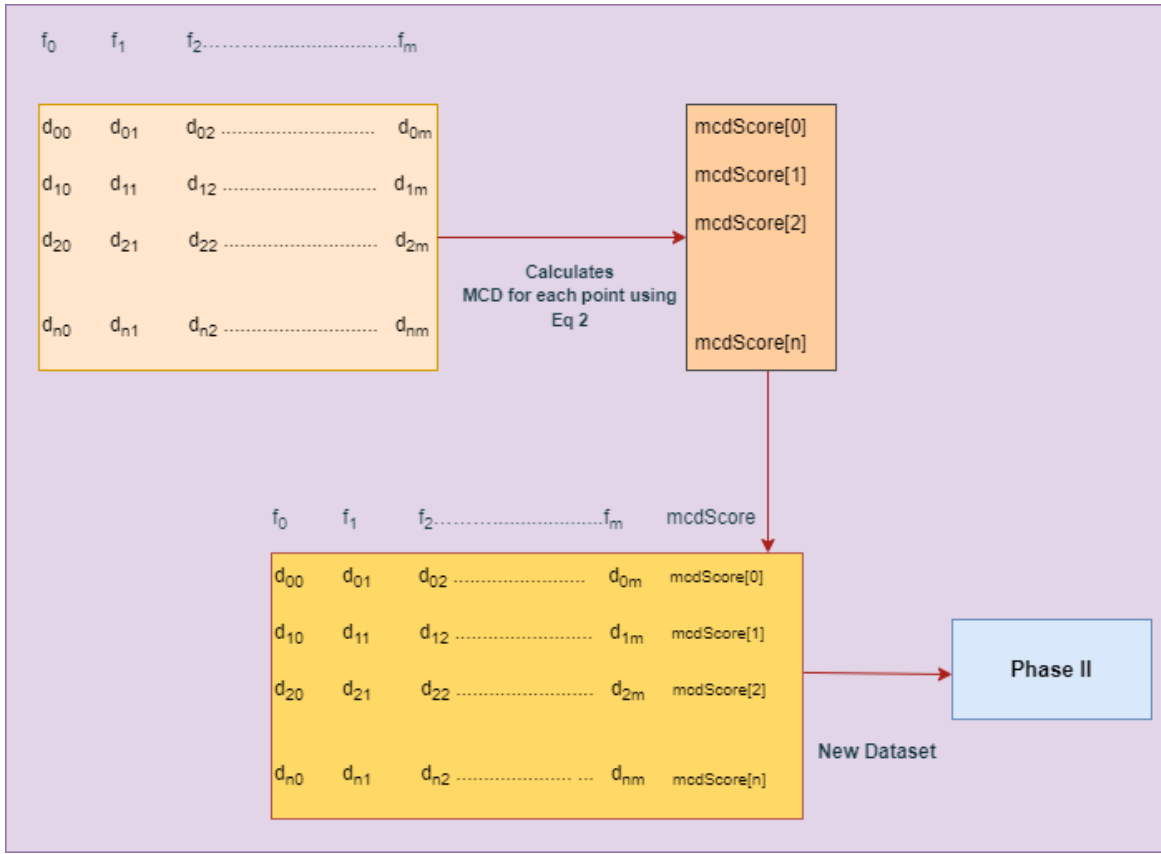


Figure 2: Forming New Dataset with MCD

Table 1: Symbols Used

Symbol	Description
n	Total number of data points
MCD^2	Squared MCD distance
μ_{MCD}	MCD Mean of the sample as defined in section 3.2
Σ_{MCD}	MCD Covariance matrix of the sample as defined in section 3.2
$mcdScore$	A feature containing mahalanobis distance corresponding to each point

set corresponding to the new dataset can be represented as $(f_1, f_2, f_3, \dots, f_m, mcdScore)$.

The performance evaluation report is generated for both the input setting to perform the comparison of results.

5 SIMULATION STUDY

This section presents the simulation setup, performance metrics, and results analysis respectively.

5.1 Simulation Setup

The experiments are performed using Google Colab on windows10 operating system. Sklearn implementation of various algorithms has been used for simulations. The language of implementation

and hardware used may affect the training time and testing time presented in section 5.3.

5.1.1 Datasets. To assess the effectiveness of the proposed approach we have employed the proposed methods to two different datasets 1) Wisconsin Breast cancer dataset (WBC) [33]. 2) Pima Indian Diabetics Dataset [33]. Description of the datasets is presented in Table 2. These are the benchmark datasets and have been used in various anomaly detection and machine learning problems by many researchers.

WBC has a total 699 instances which has real values for features for each record out of which 458 records are benign and 241 are malignant records. There are 2 class labels with WBC data. Class 2 and class 4 which we have translated class label '2' to label '0' and class

Table 2: UCI benchmark datasets used for experiments

Dataset	No of Instances	No of Classes	No of Benign Samples	No of Anomalous Samples	Percentage of Anomaly
WBC	699	2	458	241	34.47
PIMA	768	2	500	268	34.89

Algorithm 1 Proposed Approach for Anomaly Detection using improved feature set

Phase I: Forming of dataset with MCD

Input: Data corresponding to the original dataset i.e., $(x_1, x_2, x_3, \dots, x_n)$ where n is total no of data points.

Output: New dataset formed with MCD

Step 1: Select the data from the original dataset.

Step 2: calculate the MCD Mean (μ_{MCD} of the distribution) for the data.

Step 3: calculate MCD covariance matrix (Σ_{MCD}) for each point x_i .

Step 4: calculate MCD distance of each point x_i in the dataset from the remaining data distribution of the dataset using the Eq (3) and store it in $mcdScore[x_i = 1 \text{ to } n]$.

Step 5: Combine the newly obtained feature values $mcdScore$ with the data corresponding to the original dataset to create a new dataset.

Step 6: provide the new dataset as output.

Phase II: Application of Machine Learning Technique for Performance Evaluation

Input: New dataset formed with MCD.

Output: Performance evaluation report, anomalies and normal data.

Step1: Select the machine learning models and perform manual hyper parameter tuning of each machine learning algorithm.

Step 2: Operating ML algorithm with 2 different input settings.

I. Apply regular feature set which is set of relevant features from the original dataset.

II. Apply new feature set which is feature set from the new dataset obtained from step 4. This new feature set is combination of relevant features from the original dataset and MCD based mahalanobis distance i.e. $(f_1, f_2, f_3, \dots, f_m, mcdScore)$.

Step3: Generate performance evaluation report on various eight different parameters such as Accuracy, Precision, Recall and F1 score, Area under ROC, Area under PR Curve, training time and testing time in each case and compare the performance.

label '4' to '1' for the ease of implementation. Pima Indians diabetes dataset contains a total of 768 instances (rows) with eight independent parameters, namely, pregnancies, glucose, blood pressure (BP), skin thickness (ST), insulin, BMI, diabetes pedigree function and age, and one dependent parameter i.e., outcome. The dataset is primarily used for predicting the diabetes on these records. To evaluate the performance of the proposed method first we have performed the data cleaning on the dataset to avoid any NULL, duplicate, missing or irrelevant values.

5.1.2 Performance Metric. We evaluate the performance of the proposed approach by comparing eight different performance metrics

over two real-world labeled datasets discussed in section 5.1.1 we have used Accuracy, F1-score, Precision, recall, Area under the ROC Curve(AUC-ROC), Area under Precision Recall curve(AUC-PR) metrics (described from Eq (4) to Eq (11) to evaluate the performance of the proposed model. To compare the performances training and testing time have also been recorded. Although we have recorded performance of the model using various parameters listed above but for any ML model reducing the large number of false positive is important along with detection actual anomalies hence F1 score is more important metric as compared to other metrics as it also considers the effect of false positive.

True positive (TP) is the no. of correctly identified anomalies. False positive (FP) are normal points which are incorrectly identified as anomalies. Accuracy refers to the number of correctly identified anomalies over the total number of data instances. The precision refers to correctly identified true samples out of total identified anomalies by the ML Algorithm. The value of Precision will be 1 for a perfect classifier. That means high value of precision is desirable. Precision becomes 1 only when the numerator and denominator are equal in Eq (5) i.e. $TP = TP + FP$, this also means FP is zero. Recall gives a measure of how accurately our model is able to identify the relevant data. Recall is also referred as Sensitivity or True Positive Rate. Recall gives correctly identified anomalies out of the actual anomalies. Precision can also be defined as (observed present)/(predicted present) and recall can be written as (predicted present)/(Total present). Since precision reverses the conditionality in recall. The tradeoff between precision and recall can be useful in selecting a threshold for model evaluation [34]. AUC-PR defines the area under a curve, where the curve is defined by a tradeoff between precision and Recall. The Precision-Recall plot is more informative for binary classifiers in case of when the dataset is imbalanced [35]. AUC ROC stands for "Area under the Curve" of the "Receiver Operating Characteristic" curve.

The equations for all these measures is given from Eq (4) to Eq (9) as follows

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

$$\text{F1 Score} = \frac{2 * \text{precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7)$$

$$\text{TrainingTime} = \text{end}_{\text{trainingtime}} - \text{start}_{\text{trainingtime}} \quad (8)$$

$$\text{TestingTime} = \text{end}_{\text{testingtime}} - \text{start}_{\text{testingtime}} \quad (9)$$

AUC ROC is a probability curve which represents the tradeoff between True Positive Rate (TPR) and False Positive Rate (FPR) for the different values of classification threshold. The classification

Table 3: Implementation Details

S.No	ML Model	Training Size	Testing Size	Hyper parameter Values
1	GMM Classifier	80%	20%	n_components=n_classes,covariance_type='tied', max_iter=20
2	Isolation Forest	80%	20%	n_estimators=50, max_samples='auto', contamination=float(0.33), max_features=1.0
3	Logistic Regression Classifier	70%	30%	penalty = 'l1', solver = 'saga', tol = 1e-6, max_iter = int(1e6), warm_start = True, C = 0.2
4	Self-Organizing Map			alpha = 0.5*alpha, no. of iterations=100, No of clusters(k) = 2

threshold is a decision threshold. For a data point if the score of classification is greater than or equal to threshold value then point is considered as positive and if it is below than threshold value point is considered as negative in a binary classifier. Where TPR and FPR are defined as follows

$$TPR = \frac{TP}{TP + FN} \quad (10)$$

$$FPR = \frac{FP}{TP + FN} \quad (11)$$

5.1.3 Parameters Setting and Implementations. After selection of the relevant data from the dataset and preprocessing, a new dataset is created using the process described in Figure 1. To evaluate the performance of the proposed MCD Based feature selection approach we have run the machine learning models with 2 different settings as described in section 4.

We have done exhaustive experiments with different parametric and non-parametric machine learning algorithms. The models used in this study are Gaussian Mixture Model (GMM), Isolation Forest (IF), Logistic Regression (LR) and Self Organizing Map (SOM). The selection of parametric and nonparametric algorithms come from a variety of approaches including probabilistic algorithms, statistics based methods, neural networks and isolation methods. It is worth noting that these approaches require exhaustive parameter tuning. Parameter tuning has been done manually for the implementation and best parameters are considered in our implementation. The hyper parameters used and other implementation details are listed in Table 3.

5.2 Simulation Results

In this section simulation study of the proposed method has been presented using different machine learning models. To perform GMM clustering no. of components (i.e., no of clusters) needs to be provided as input to the Gaussian Mixture Model. To determine the correct no of clusters which are suitable for the given data we have first plotted the graph between different no of clusters and corresponding silhouette score. Silhouette score is an indication of the quality of clustering is shown for two input settings in Figure 3 (a) and Figure 3 (b) respectively. It can be observed that the value of silhouette score is maximum when value of number of clusters is taken as two in both the settings so two is also taken as no. of component for the implementation of GMM. It can also be observed that the silhouette score is also clearly higher when GMM Clustering is performed with new feature. The value of silhouette

score is 0.60 and 0.64 for regular feature set and improved feature set respectively over WBC dataset.

GMM method has been implemented as a Classifier in the literature by various researchers [36]. After identifying the correct no of component further we have applied Gaussian Mixture model as a classifier in our work. Figure 4 (a) and (b) presents the AUC ROC curve obtained for GMM Classification in both settings respectively over WBC dataset. The Area under the Curve (AUC) is the measure of the ability of a binary classifier to distinguish between classes. Greater the area under the curve better the performance of the classifier. There is an increase in area under curve from 0.9891 to 0.9927 in setting II which can be clearly observed from the figure.

Area under the curve for GMM Classifier over PIMA dataset is represented by Figure 5 (a) and Figure 5 (b) respectively using two input feature set. It is clear from the graph that the AUC ROC has improved with improved feature set i.e., from .4138 to .4895. The AUC ROC value over PIMA diabetic dataset is considerably low as compared to WBC dataset. This can be explained by the characteristics of the dataset. But there is an increase from setting I in the AUC-ROC value for GMM when using input setting II.

Figure 6 (a) and (b) represents the Area under ROC Curve for isolation forest algorithm in two settings respectively over WBC. Same is shown in Figure 7(a) and (b) over PIMA diabetic datasets. Here graphs and simulations results are obtained through 10-fold cross validation techniques. The AUC ROC value is improved from 0.96 to 0.98 over WBC and from 0.62 to 0.68 using setting II as shown in the Figure 6 and Figure 7 respectively.

As depicted in Figure 8 and Figure 9 the value of AUC-PR is also increased when new feature set is applied on Isolation Forest over WBC from 0.92 to 0.95 and over PIMA from 0.51 to 0.54 respectively. Although the value of AUC-PR over PIMA dataset is not encouraging in itself. This relates to the very nature of the dataset but here again new feature set helps to improve the result by 3%.

As shown in Figure 10(a) and (b) the Area under ROC Curve for Logistic regression algorithm has improved by 1% from .98 to .99 when using improved feature set over WBC dataset. Similarly, AUC ROC for Logistic regression has improvement in case of PIMA dataset shown in Figure 11 (a) and (b). The improvement in the AUC PR is also visible from Figure 12 and Figure 13 over WBC and PIMA dataset respectively.

From the above results it is observed that ML models when executed with the proposed methodology outperforms the ML models when executed using regular feature for GMM Classifier, Isolation

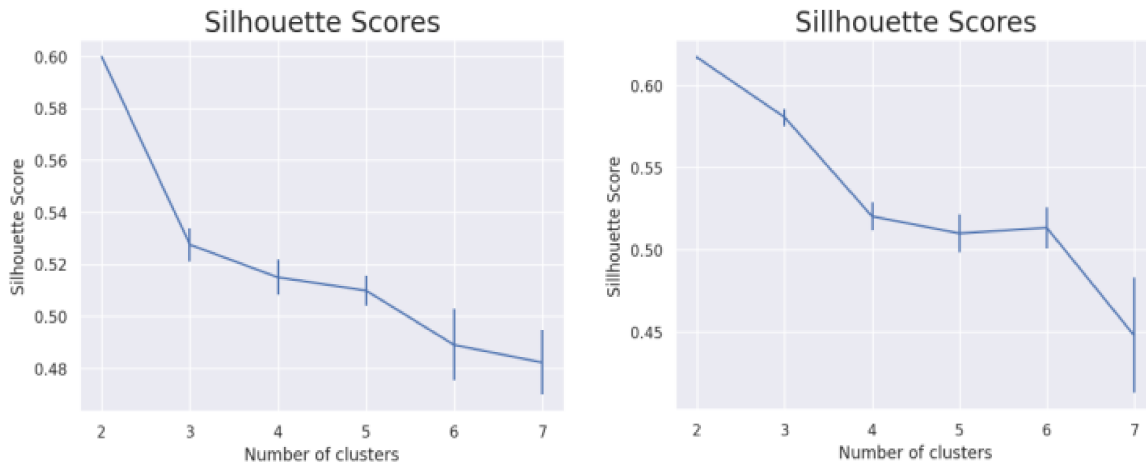


Figure 3: Silhouette Score for GMM over WBC dataset using a) setting I b) setting II

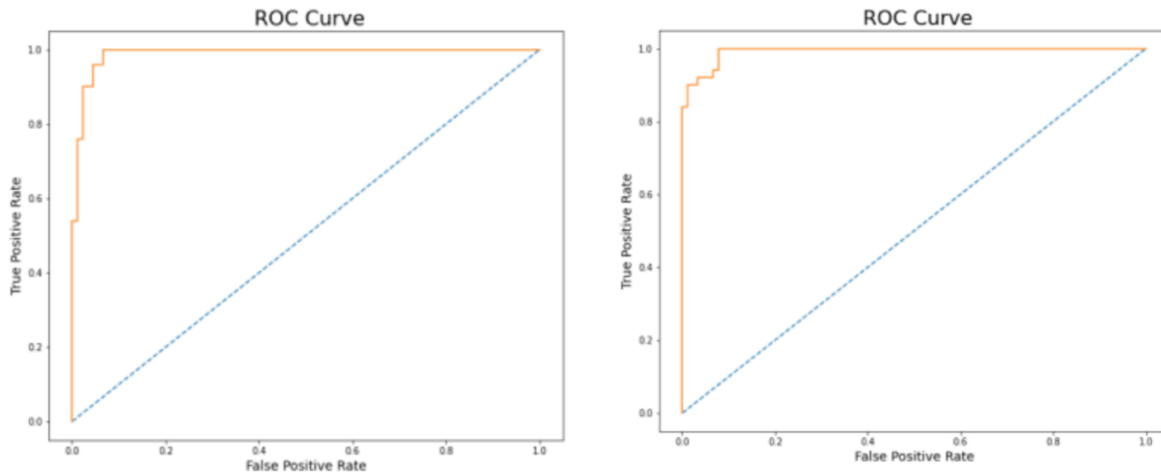


Figure 4: AUC ROC Curves for GMM Classification over WBC dataset using a) setting I b) setting II

Forest and Logistic Regression Classifier. The Pima Indians diabetes dataset has the comparatively low AUC ROC and AUC PR Score for both types of input feature settings which indicates that the machine learning algorithm find difficultly in classifying the dataset.

Self-Organization Map (SOM) is a neural network-based clustering technique which can be used to identify anomalies. SOM algorithm has been implemented with no. of clusters as two and Initial learning rate has been chosen as 0.5. 100 iterations for the self-organizing map have been performed and the learning rate is updated by 0.5 in each iteration. The total time required by SOM is also recorded in both the cases.

As shown in Table 4 although there is very little improvement in case of WBC dataset and no improvement in case of PIMA dataset for SOM in terms of accuracy, precision, recall and F1 Score but the total time required is reduced significantly in SOM with input setting II. Which means that the learning time required for SOM reduces significantly when it is executed with the additional vital

information of MCD distance. This proves the superiority of improved feature selection based on time required to train the model.

With these observations, it can be concluded that using the proposed Minimum Covariance Determinant feature based proposed approach is more effective than using original features of the dataset.

5.3 Results Summary

Table 4 and Table 5 show the results obtained for SOM using regular and improved feature set. Table 6 and Table 7 show the accuracy, precision, recall and F1 score respectively for GMM for both WBC and Pima datasets respectively.

Figure 14 and Figure 15 depict the outcome over WBC and PIMA dataset respectively, in the form of a graph. The proposed method demonstrated a better accuracy, precision, recall and F1 Score along with higher values of AUC-ROC and AUC-PR with comparable training and testing time. In most of the cases indicates reasonably better performance of the proposed feature selection method. The

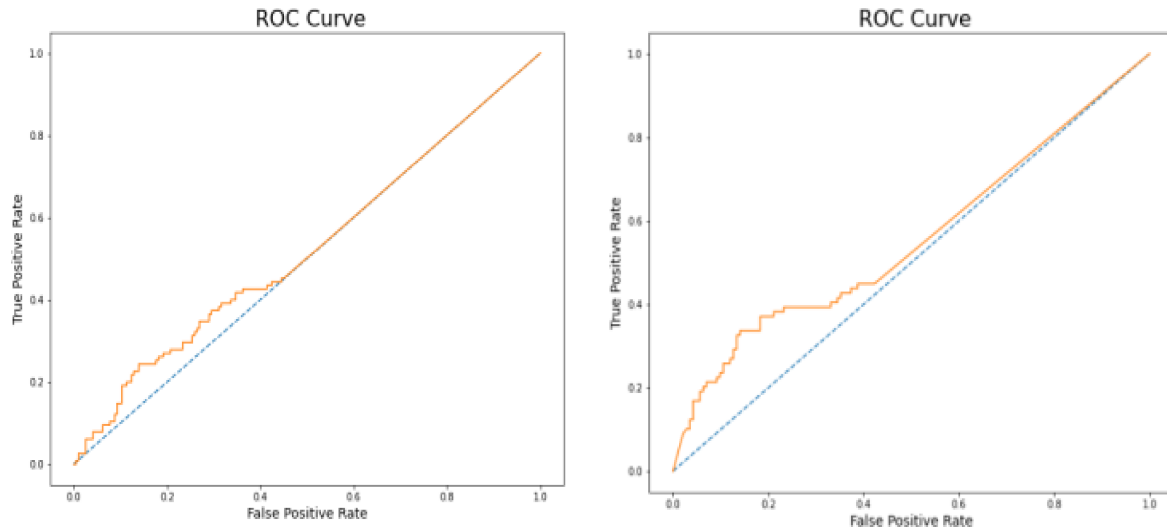


Figure 5: AUC ROC Curves for GMM Classification over PIMA dataset using a) setting I b) setting II

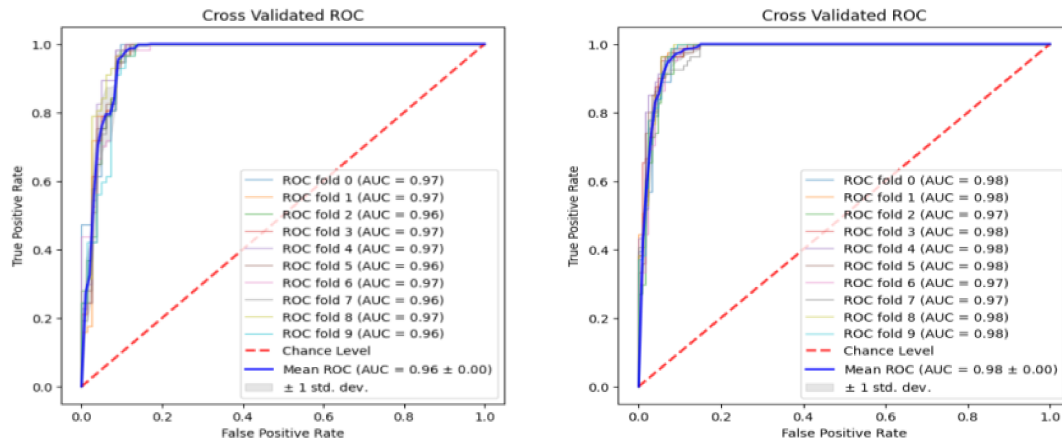


Figure 6: AUC ROC Curves for Isolation Forest Algorithm over WBC dataset using a) setting I b) setting II

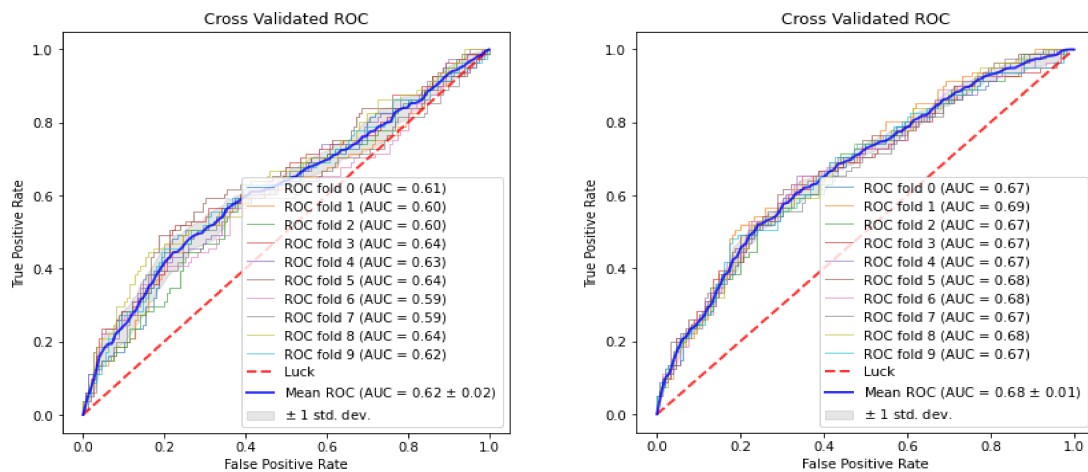


Figure 7: AUC ROC Curves for Isolation Forest Algorithm over PIMA dataset using a) setting I b) setting II

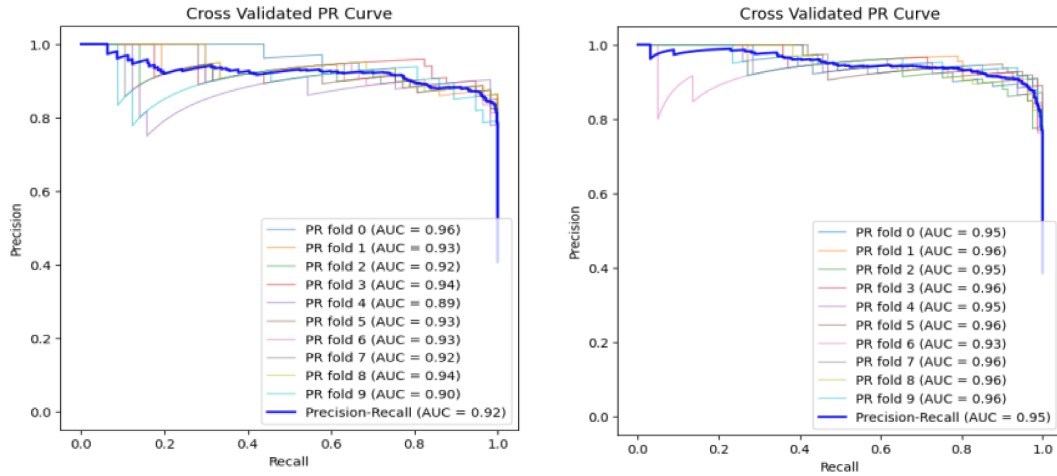


Figure 8: AUC PR Curves for Isolation Forest Algorithm over WBC dataset using a) setting I b) setting II

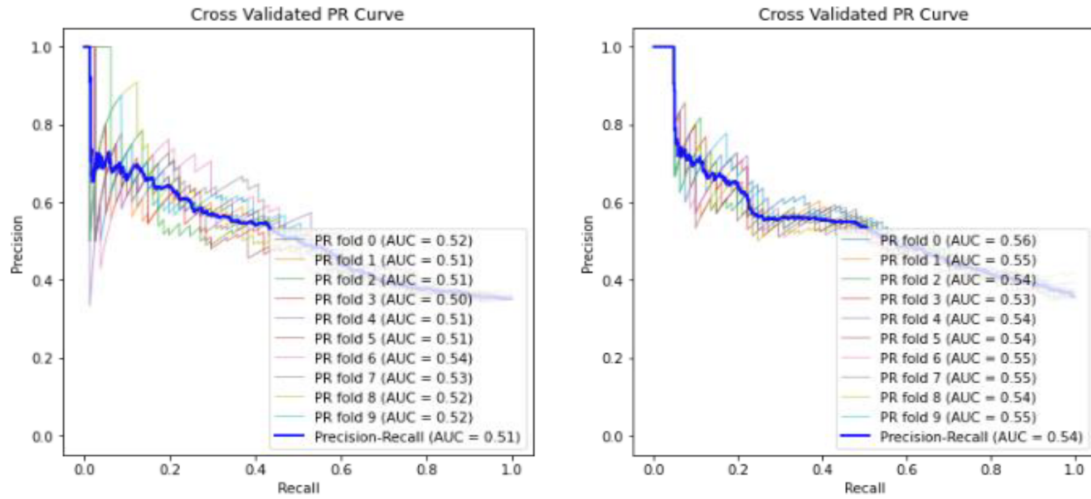


Figure 9: AUC PR Curves for Isolation Forest Algorithm over PIMA dataset using a) setting I b) setting II

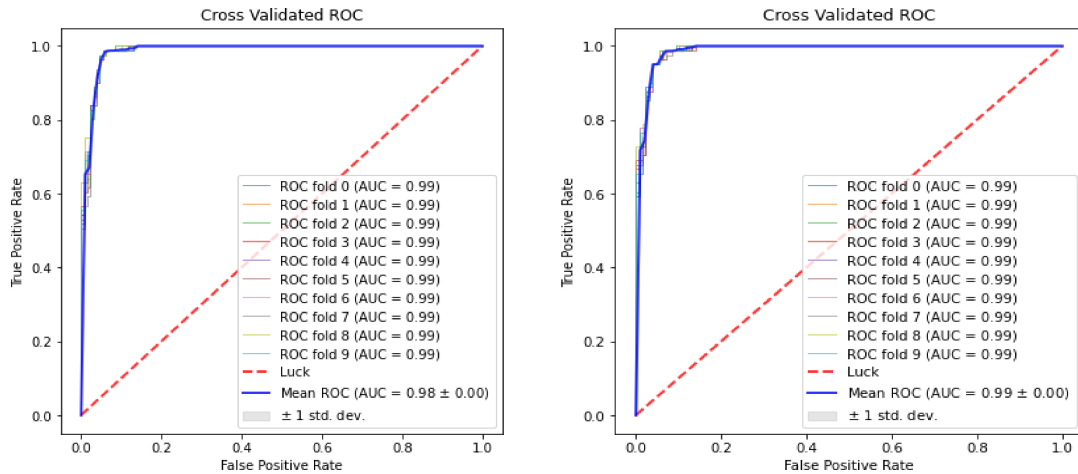


Figure 10: AUC ROC Curves for Logistic regression over WBC dataset using a) setting I b) setting II

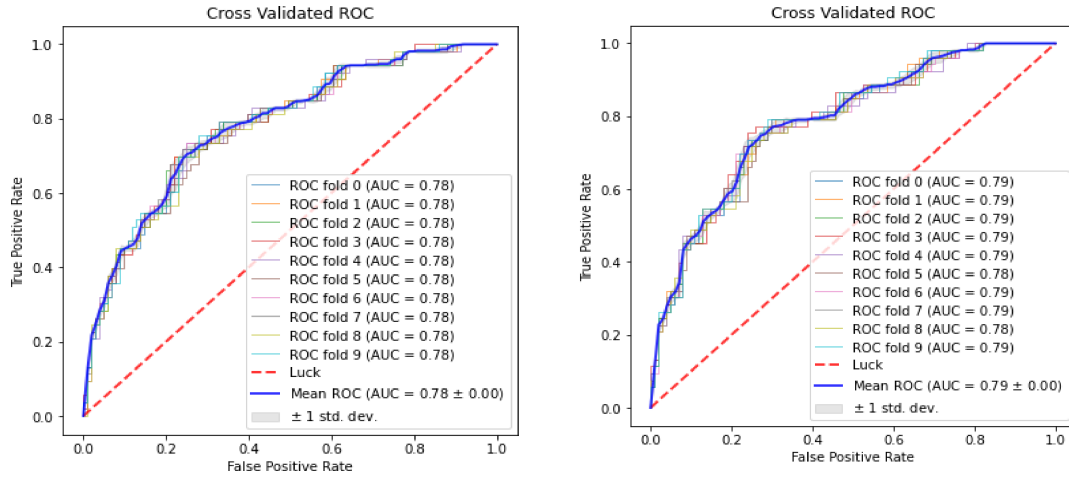


Figure 11: AUC ROC Curves for Logistic regression over PIMA using a) setting I b) setting II

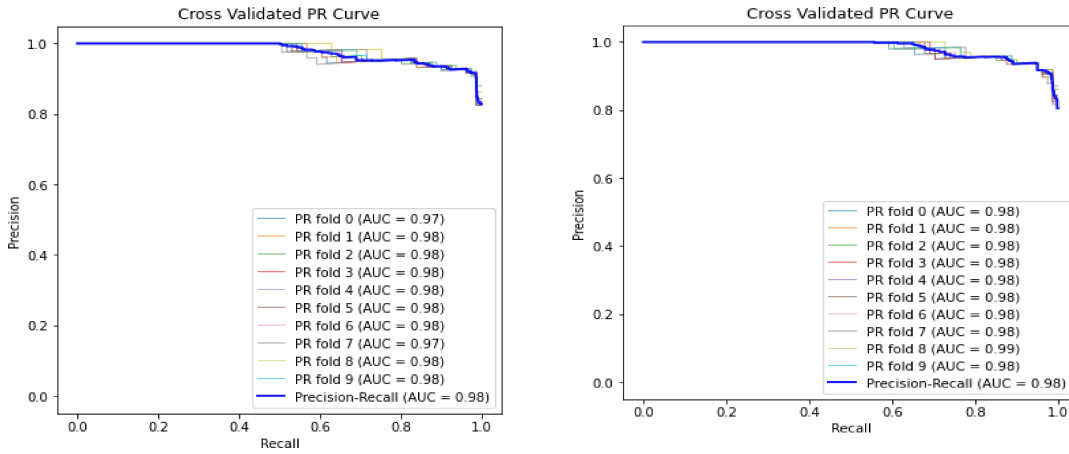


Figure 12: AUC PR Curves for Logistic regression over WBC using a) setting I b) setting II

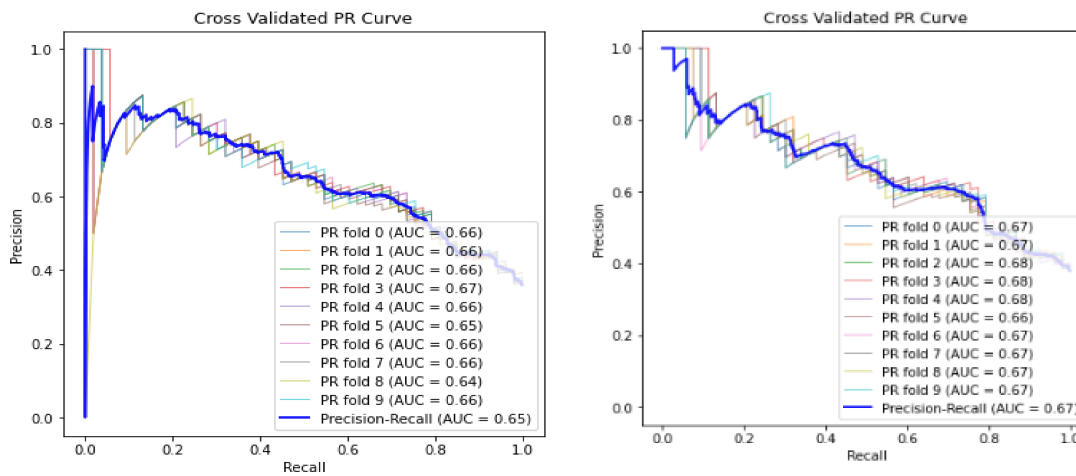


Figure 13: AUC PR Curves for Logistic regression over PIMA using a) setting I b) setting II

Table 4: Results of SOM Clustering on WBC Dataset

SOM Clustering on WBC Dataset		
	With regular method	With proposed Method
Accuracy	95.88	96.06
Precision	95.65	95.67
Recall	92.14	92.67
F1 Score	93.86	94.14
Total Time (sec)	2.73	1.193

Table 5: Results of SOM Clustering on PIMA Diabetic Dataset

SOM on PIMA Dataset		
	With Regular method	With proposed Method
Accuracy	66.44	66.44
Precision	51.21	51.21
Recall	30.17	30.17
F1 Score	37.95	37.95
Total Time (sec)	5.26	1.82

Table 6: Results of classifications over WBC dataset

Performance Measures	GMM Classifier		Isolation Forest		Logistic Regression	
	With Regular method	with proposed Method	With Regular method	with proposed Method	With Regular method	with proposed Method
Accuracy	94.28	95.23	91.14	91.9	94.61	95.23
Precision	93.75	94.02	91.05	90.64	94.26	94.81
Recall	90	91.3	73.09	74.49	57.02	58.56
F1 Score	91.83	92.64	91.14	91.19	94.61	95.23
Training Time (in secs)	0.0603	0.1489	0.168	0.09528	0.3963	1.5956
Testing Time(in secs)	0.0337	0.0033	0.0728	0.04508	0.06934	0.1515
AUC PR Score	1	1	0.9237	0.9507	0.9767	0.9821
AUC ROC Value	0.99	0.99	0.9638	0.9764	0.9843	0.9862

Table 7: Results of classifications over PIMA dataset

Performance Measures	GMM Classifier		Isolation Forest		Logistic Regression	
	With Regular method	with proposed Method	With Regular method	with proposed Method	With Regular method	with proposed Method
Accuracy	51.29	52.81	61.81	65.62	73.96	74.15
Precision	37.41	40	46.43	65.62	71.22	71.6
Recall	45.21	44.94	58.42	60.77	67.03	65.61
F1 Score	40.94	42.32	51.15	65.62	73.96	74.15
Training Time (in secs)	0.3037	0.03822	0.9244	1.01	0.1214	0.0113
Testing Time (in secs)	0.0031	0.0028	0.4252	0.4623	0.0102	0.0128
AUC PR Score	0.52	0.56	0.6172	0.675	0.7833	0.7893
AUC ROC Score	0.4138	0.4895	0.5091	0.5704	0.6513	0.6674

results clearly indicate that our proposed news feature set is capable of improving the detection of outliers with multiple attributes. It is evident that the proposed approach is useful and beneficial with

any type of machine learning algorithm and help to improve the performance of the model on various performance measures with comparable time requirement.

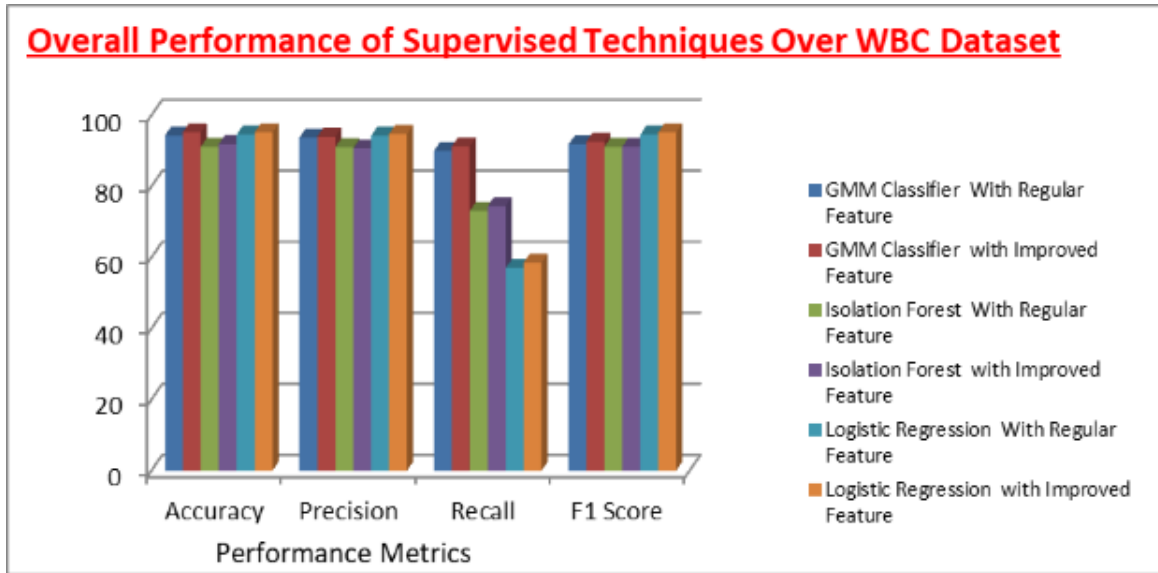


Figure 14: Overall Performance comparison of the proposed approach over WBC Dataset

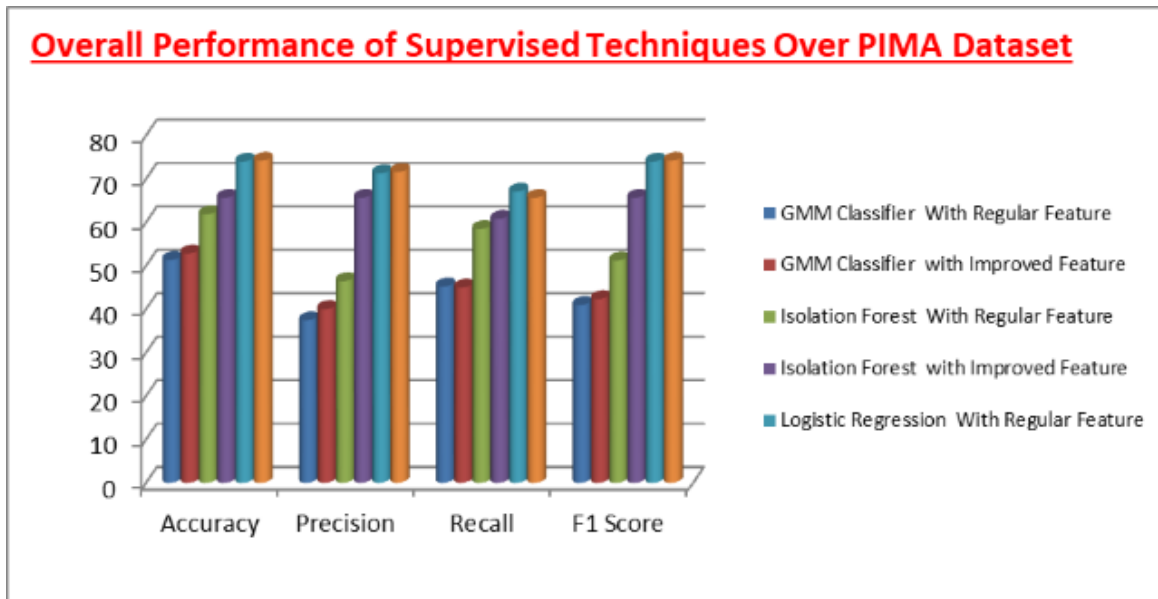


Figure 15: Overall Performance comparison of the proposed approach over PIMA Dataset

6 CONCLUSION AND FUTURE WORK

Data generated by various sources is susceptible to anomalies. Use of Distance information can be effective for detection of anomalies. Our proposed approach utilizes the MCD based mahalanobis distance for improving the detection of anomalies in combination with various machine learning models. We explored and demonstrated the possibilities of fusion of MCD based mahalanobis distance with machine learning algorithms for anomaly detection. We proved

the efficacy of our proposed approach on various machine learning models: Logistic Regression classifier, Gaussian mixture model, Isolation Forest algorithms and self-organizing map. The machine learning models performs better using the proposed approach with four machine learning models which are examined in this work. As of its limiting factors are

Concerned to the fact that the calculation of MCD is time consuming and when the dataset is large it may increase the time and memory requirements.

It can be established that our proposed approach performed consistently better on various machine learning models. This shows the applicability of the proposed approach hereby use of MCD with the machine learning models is deemed effective. In future we plan to use MCD based mahalanobis distance in combination with deep learning models for anomaly detection in multivariate data.

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